Particle-sector closure from the TGP substrate

Charged-lepton masses, Koide $K = 2/3$, the Cabibbo angle,
and the sector-specificity of Koide across quarks and neutrinos

Mateusz Serafin

Independent researcher

environmenstefan@gmail.com

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Abstract

Building on the *Theory of Generated Space* (TGP) introduced in [1], we apply the substrate-field framework to the charged-lepton sector. Starting from the same soliton ordinary differential equation (ODE) that fixes the substrate profile in the core theory, we derive, without any new degree of freedom: (i) the lepton mass law $m_{\rm phys} = c_M\, A_{\rm tail}^4$ with $k = 4$ uniquely selected by $k = 2(d-1)/(d-2)$ for $d = 3$ and verified to $(A_\mu/A_e)^4 = 206.74$ against the PDG ratio 206.768 at the 0.013% level; (ii) the integer generation count $N = 3$ as a metric-singularity barrier, $g_0^{\rm crit}(d = 3, \alpha = 1) = 2.206$, with $g_0^\tau < g_0^{\rm crit}$ and $g_0^{(4)} > g_0^{\rm crit}$; (iii) the Koide relation $K_{\rm K} = 2/3$ as a theorem of the soliton ODE, with two independent extractions of g_0^τ (one from the mass ratio, one from $K_{\rm K} = 2/3$) agreeing to 0.036%; the perturbative coefficient $\alpha_3 = \pi^2/128 + P_{\rm cos}$ is decomposed analytically into a proven part and a residual primitive $P_{\rm cos}$ with 40-digit numerical value, which is shown by PSLQ on a 24-element polylog/Clausen basis at $\max\text{coef} = 10^{14}$ to admit no relation and is therefore promoted to a TGP-named constant; (iv) the Cabibbo angle via a $\mathbb{Z}_3$ self-energy subtraction on $\mathrm{GL}(3, \mathbb{F}_2)$, reducing the PDG tension from 4.8σ at zero order to 0.75σ. We then show that the Koide relation is *not* universal: the quark Koide ratios $K_u = 0.8746$ and $K_d = 0.7398$ are invariant under 2-loop QCD running of γ_m (the flavour-blind multiplicative factor cancels identically between numerator and denominator), and the neutrino Brannen-$\sqrt{2}$ ansatz is falsified by the combined oscillation Δm^2 data with NuFit 5.3 inputs (minimum Brannen residual 4.74, not consistent at any phase angle; $\max K_\nu^{\rm NO} = 0.553$ below 2/3 by +0.114). Koide $K_{\rm K} = 2/3$ is therefore established as a charged-lepton signature of the soliton ODE, not a universal law.

Keywords: charged leptons, Koide relation, Cabibbo angle, lepton mass formula, soliton, generated space, TGP, scalar field, Brannen, $\mathrm{GL}(3, \mathbb{F}_2)$, neutrino mass, PSLQ.

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1 Introduction

The charged-lepton mass ratios $m_e : m_\mu : m_\tau \approx 1 : 206.768 : 3477.2$ have no explanation in the Standard Model: each mass is a free Yukawa coupling, and the remarkable Koide relation [3, 4, 5]

$$K_{\rm K} \equiv \frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} = \frac{2}{3} \quad (\text{PDG } 1.5000137\ldots) \tag{1}$$

holds experimentally to better than one part in 10^5 without any first-principles derivation in the SM. Attempts to explain $K_{\rm K} = 2/3$ through symmetry groups, discrete family symmetries or fixed-point dynamics have all either added additional free parameters or failed to recover the numerical value without fine-tuning.

Within the *Theory of Generated Space* (TGP) [1], the fundamental degree of freedom is a scalar substrate field Φ; a single non-linear potential $U(\Phi)$ with $U'(\Phi_0) = 0$ admits a localised soliton solution whose tail decays as $\Phi(r) \sim \Phi_0 - A_{\rm tail}\, e^{-r}/r$. The present paper demonstrates that the charged-lepton sector is determined by this soliton ODE essentially completely, with a single overall mass scale c_M as the only residual input:

- the mass scaling law $m_{\rm phys} = c_M\, A_{\rm tail}^4$ with $k = 4$ uniquely selected in $d = 3$ (Sec. 3),
- the number of generations $N = 3$ from a metric-singularity barrier in the soliton core (Sec. 4),
- the Koide relation $K_{\rm K} = 2/3$ as a theorem of the ODE perturbative expansion, with two independent extractions of the third eigenvalue agreeing to 0.036% (Sec. 5),
- the Cabibbo angle via a $\mathbb{Z}_3$ self-energy subtraction on $\mathrm{GL}(3, \mathbb{F}_2)$, reducing PDG tension from 4.8σ to 0.75σ (Sec. 6).

Crucially, we then show that $K_{\rm K} = 2/3$ is *not* universal. The quark Koide ratios are RG-invariant under 2-loop QCD running and do not equal 2/3 (Sec. 7); the neutrino Brannen-$\sqrt{2}$ ansatz is falsified by oscillation Δm^2 data. Koide $K_{\rm K} = 2/3$ is therefore a charged-lepton signature of the soliton dynamics fixed by [1], not an accidental empirical fact to be explained at all costs.

Structure. Section 2 fixes notation for the substrate soliton inherited from the TGP core. Section 3 derives the $k = 4$ mass law. Section 4 derives $N = 3$ from the metric singularity. Section 5 states and proves $K_{\rm K} = 2/3$ as a theorem of the soliton perturbation theory. Section 6 derives the Cabibbo correction. Section 7 establishes Koide sector-specificity. Section 8 collects the explicit open problems (principally: analytic identification of the residual TGP primitive $P_{\rm cos}$). Section 9 summarises.

2 The substrate soliton

Throughout this paper we use the substrate conventions of the TGP core [1]. We summarise here only the content strictly required to follow Secs. 3–6.

2.1 Lagrangian and field equation

The dimensionless substrate field $g(\mathbf{r}) \equiv \Phi/\Phi_0$ obeys the Lagrangian density

$$\mathcal{L}[g] = \tfrac{1}{2}\, g^{2\alpha}\, (\nabla g)^2 + U(g), \qquad U(g) = \tfrac{1}{3} g^3 - \tfrac{1}{4} g^4, \tag{2}$$

where the kinetic exponent $\alpha = 1$ is fixed by the algebraic uniqueness theorem of the core paper [1]. The potential U has $U'(g) = g^2(1-g)$ with critical points $g = 0$ (the absolute reference state $\aleph_0$) and $g = 1$ (the vacuum). A localised, spherically symmetric soliton solution $g(r)$ obeys the Euler–Lagrange equation

$$\boxed{g''(r) + \frac{2\,[g'(r)]^2}{g(r)} + \frac{2}{r}\, g'(r) = 1 - g(r),} \tag{3}$$

with regularity at $r = 0$ and asymptotic boundary condition $g(r) \to 1$ as $r \to \infty$. The one-parameter family of solutions is labelled by the core value $g_0 \equiv g(0)$. The tail behaviour is fixed by the linearisation around $g = 1$:

$$g(r) \;\sim\; 1 \;-\; A_{\text{tail}}\,\frac{e^{-r}}{r}, \qquad r \to \infty, \tag{4}$$

so that a soliton is specified either by g_0 or by the tail amplitude A_{tail}; the two are related by a monotone bijection on the regular branch (proved in [1] and numerically verified for $g_0 \in (0, g_0^{\text{crit}})$ in [2]).

2.2 Conservation law

Integrating Eq. (3) against $r^{2(d-1)}$ in spatial dimension d yields the universal conservation law

$$\left[r^{2(d-1)}\,q(r)\right]' \;=\; r^{2(d-1)}\,U'(g(r)), \tag{5}$$

where $q = g^{2\alpha} g'$ is the radial flux. This identity holds for all (α, d) and will be used in Sec. 3 to prove the virial mass formula.

2.3 Metric-singularity barrier and g_0^{crit}

For $d = 3$ and $\alpha = 1$ there exists a critical core value $g_0^{\text{crit}}(d = 3, \alpha = 1) = 2.206$ above which $g(r) \to 0$ at some finite $r = r_* > 0$, triggering a singularity of the effective metric $g_{ij} = e^{+2U/c_0^2}\delta_{ij}$ of the core paper (the metric degenerates when $g \to 0$ in the soliton core). Soliton branches with $g_0 > g_0^{\text{crit}}$ are therefore dynamically forbidden; this barrier will be used in Sec. 4 to derive $N = 3$.

Numerical values of g_0^{crit} for illustrative (d, α) pairs are reported in Table 1 and cross-checked against the conservation law (5) via the universal identity $(r^{2(d-1)}q)' = r^{2(d-1)}U'$, which passes at 4-digit precision for $(d, \alpha) \in \{(2, 1), (3, 1), (3, 1/\varphi), (3, 0.5), (3, 0.8), (3, 1.0)\}$ (full table in the companion repository [2]).

Table 1: Critical core value g_0^{crit} of the substrate soliton for representative (d, α). The $d = 3, \alpha = 1$ entry is the physically relevant one.

d	α	g_0^{crit}	Context
1	any	4/3	algebraic theorem, U-independent of α
3	0.50	2.618	reference $K = g$ kinetic
3	0.62	2.487	$1/\varphi$ (golden)
3	0.80	2.332	intermediate
3	1.00	**2.206**	**substrate (physical)**

3 Charged-lepton mass law: $m_{\text{phys}} = c_M\, A_{\text{tail}}^4$

We now derive the charged-lepton mass law directly from the soliton ODE (3), with the integer exponent $k = 4$ fixed by dimensional algebra in $d = 3$.

3.1 Virial identity and the K^2 mechanism

Define the kinetic and potential integrals

$$K \;\equiv\; \int_0^{R_{\max}} \tfrac{1}{2}\, g^{2\alpha}(g')^2\, r^2\, dr, \qquad |V| \;\equiv\; \left|\int_0^{R_{\max}} U(g)\, r^2\, dr\right|. \tag{6}$$

The zero-mode virial identity $E^{(2)} = 0$ of the soliton yields the *exact* relation [2]

$$K \;=\; C_T \cdot A_{\text{tail}}^2, \qquad C_T \;=\; R_{\max}/4 + C_{\text{core}}, \tag{7}$$

with $C_{\text{core}}/A_{\text{tail}}^2 \approx 1.09$ a topological constant (fluctuation $< 1.5\%$ at $d = 3, \alpha = 1$).

Proposition 1 (Virial mass formula). *The physical mass of a soliton of tail amplitude A_{tail} is*

$$m_{\text{phys}} = c\,K^2, \tag{8}$$

for a universal constant c determined by the substrate normalisation of the core paper.

A detailed derivation of Eq. (8) by integrating the stress-energy density against the effective metric is given in [1]; the script `research/mass_scaling_k4/r5_k_squared_mechanism.py` verifies Eq. (8) on the soliton branch for $R_{\max} \in [40, 100]$ with *7/7 PASS* against the ratio prediction $(K_\mu/K_e)^2 = (A_{\text{tail},\mu}/A_{\text{tail},e})^4$.

Theorem 2 (Integer-k selection in $d = 3$). *Under the virial mechanism of Prop. 1, the mass-ratio scaling law is*

$$\frac{m_{\text{phys},i}}{m_{\text{phys},j}} = \left(\frac{A_{\text{tail},i}}{A_{\text{tail},j}}\right)^k, \qquad k = \frac{2(d-1)}{d-2}. \tag{9}$$

For $d = 3$, $k = 4$ is the unique integer value. (For $d = 2$ the quantity diverges; for $d = 4$, $k = 3$; for $d = 5$, $k = 2.667$.)

Sketch. The virial integrals $K, |V|$ both scale as A^2_{tail} up to the $R_{\max}/4$ term, which cancels in the ratio. Hence $m_{\text{phys}} \propto K^2 \propto A^4_{\text{tail}}$ in $d = 3$. The exponent $k = 2(d-1)/(d-2)$ comes from the radial measure r^{d-1} in the kinetic integral weighted against the asymptotic $(g')^2 \sim (e^{-r}/r)^2(\ldots)$ fall-off; only $d = 3$ selects $k \in \mathbb{Z}$. A detailed derivation is in [2], `research/mass_scaling_k4/r5_virial_mass_derivation.py`. □

3.2 Numerical verification against PDG

For the three charged-lepton branches with core values $\{g_0^e, g_0^\mu, g_0^\tau\}$ (see Sec. 5) the tail amplitudes $\{A^e_{\text{tail}}, A^\mu_{\text{tail}}, A^\tau_{\text{tail}}\}$ are obtained numerically from Eq. (3) and converted to mass ratios via Eq. (9).

Table 2: PDG mass ratios vs the $k = 4$ law. The exponent $k_{\text{eff}} \equiv \ln(m_i/m_j)/\ln(A_{\text{tail},i}/A_{\text{tail},j})$ is extracted numerically.

Quantity	TGP value	PDG value	Relative error
$(A_{\text{tail},\mu}/A_{\text{tail},e})^4$	206.74	206.768	$1.3 \cdot 10^{-4}$
k_{eff}	4.0008	4 (integer)	$2 \cdot 10^{-4}$
$(A_{\text{tail},\tau}/A_{\text{tail},e})^4$	3477	3477.23	$7 \cdot 10^{-5}$

The $k = 3$ and $k = 5$ hypotheses give 55 and 784 respectively for m_μ/m_e, vs observed 206.768, confirming the $k = 4$ selection unambiguously.

3.3 Why a perturbative derivation of $k = 4$ fails

A naïve perturbative derivation of $k = 4$ through the action expansion *fails*: the third-order piece $E^{(3)}$ does not vanish on the soliton background, and its ratio $|E^{(3)}/E^{(4)}| \sim A^{-0.9}_{\text{tail}}$ *diverges* for small solitons [2]. The correct derivation is *non-perturbative* via Prop. 1: K and $|V|$ are close in absolute value ($K/|V| \approx 1.013$) but their A^2_{tail}-scaling is *individually* clean, and the ratio (8) is exact. This subtlety is the reason the law was stated as an empirical observation for two decades before the virial mechanism was identified.

4 Three generations: $N = 3$ from the metric barrier

4.1 The barrier argument

By Sec. 2, the soliton branch is restricted to $g_0 \in (0, g_0^{\text{crit}})$ with $g_0^{\text{crit}} = 2.206$ for $d = 3, \alpha = 1$. The effective metric $g_{ij} = e^{+2U/c_0^2}\delta_{ij}$ is singular wherever $g(r) \to 0$, and g does reach zero at a

finite radius for $g_0 > g_0^{\rm crit}$. A fourth charged-lepton generation would require $g_0^{(4)} > g_0^{\tau}$; the mass pattern dictated by Eq. (9) and the PDG ratios gives

$$g_0^e \approx 1.0, \quad g_0^\mu \approx 1.5, \quad g_0^\tau \approx 1.73, \quad g_0^{(4)} \gtrsim 2.2, \tag{10}$$

so that $g_0^{(4)}$ would exceed $g_0^{\rm crit} = 2.206$ and trigger the metric singularity. We therefore have:

Theorem 3 (No fourth charged-lepton generation)**.** *In $d = 3$ with substrate kinetic exponent $\alpha = 1$, at most three charged-lepton solitons with regular effective metric exist. A fourth branch would require $g_0 > g_0^{\rm crit}$ and triggers a metric singularity that renders the soliton dynamically inadmissible.*

4.2 Mass divergence near the barrier

The soliton mass diverges as $g_0 \to g_0^{\rm crit}$ from below: numerically, $dm/dg_0 \to \infty$ at $g_0 = g_0^{\rm crit}$ [2]. This provides a *hard* kinematic limit, not merely a dynamical preference: there is no finite continuation beyond $g_0 = g_0^{\rm crit}$. The script `research/why_n3/r3_n3_from_barrier.py` exhibits the divergence explicitly.

4.3 Cross-check: α-dependence of the barrier

Table 1 shows that $g_0^{\rm crit}(d = 3, \alpha)$ is monotonically decreasing in α. The critical value of α at which the barrier first admits exactly three generations is $\alpha_* \approx 0.882$, and the substrate choice $\alpha = 1$ sits just above this threshold with a 3% deficit, placing the physical substrate very near the $N = 2 \to N = 3$ boundary (a near-critical statement not explored further in this paper). The scripts `r3_alpha_scan.py` and `r3_g0crit_analytical.py` reproduce this result.

5 The Koide relation $K_{\rm K} = 2/3$

5.1 Algebraic reduction to the Brannen ratio

Define $s_i \equiv \sqrt{m_i}$ for $i \in \{e, \mu, \tau\}$ and the Brannen parametrisation [5]

$$s_i \;=\; a\big[\,1 + B\,\cos(\theta + 2\pi i/3)\,\big], \qquad B \equiv b/a, \tag{11}$$

for three real numbers $\{s_e, s_\mu, s_\tau\}$. The phases $\{0, 2\pi/3, 4\pi/3\}$ are not an ansatz but a theorem of the discrete Fourier transform on $\mathbb{Z}_3$: any three real positive numbers admit a unique representation of the form (11) with equispaced phases.

Proposition 4 (Algebraic Koide identity)**.** *For any three real positive $\{s_i\}$ parametrised as* (11)*,*

$$K_{\rm K} \;=\; \frac{\sum_i s_i^2}{\left(\sum_i s_i\right)^2} \;=\; \frac{1 + B^2/2}{N}, \qquad N = 3. \tag{12}$$

In particular, $K_{\rm K} = 2/N$ is equivalent to $B^2 = 2$, and for $N = 3$ equivalent to $B = \sqrt{2}$.

Proof. Direct substitution of (11) into the definition and using $\sum_i \cos(\theta + 2\pi i/3) = 0$ and $\sum_i \cos^2(\theta + 2\pi i/3) = 3/2$. □

Numerically, PDG lepton masses yield $B_{\rm num} = 1.41421\ldots$ with $|B_{\rm num} - \sqrt{2}| < 10^{-6}$, and correspondingly $K_{\rm K} = 0.666671(\ldots)$. Proving $B = \sqrt{2}$ *analytically* from the soliton ODE would therefore promote $K_{\rm K} = 2/3$ from a numerical coincidence to a theorem of the substrate theory, which is the programme of the next subsections.

5.2 Perturbative expansion of the soliton ODE

Let $\delta \equiv g_0 - 1$ measure the displacement of the soliton core from the vacuum. For small δ, substitute the ansatz

$$A_{\text{tail}}(g_0) \;=\; c_1\,\delta\,\big[1 + \alpha_2\,\delta + \alpha_3\,\delta^2 + O(\delta^3)\big], \tag{13}$$

into the soliton ODE (3) and extract the coefficients order by order by matched asymptotics between the core region and the exponential tail.

Theorem 5 (Analytic coefficients c_1 and α_2)**.** *The leading perturbative coefficient is*

$$c_1 \;=\; 1 - \frac{\ln 3}{4} \;\approx\; 0.72534693, \tag{14}$$

and consequently

$$\alpha_2 \;=\; c_1/2 \;=\; \tfrac{1}{2} - \tfrac{\ln 3}{8}. \tag{15}$$

Sketch. Frullani-type integral identities applied to the $O(\delta)$ equation yield $c_1 = \int_0^1 (1-u^2)\,du/(u+1) - \int_0^1 \ldots$, which sums telescopically to $1 - \ln 3/4$. For α_2, the $O(\delta^2)$ equation splits into a trivial piece and a c_1-proportional piece; the coefficient follows by virial balance. Full computation in the scripts `r6_c1_closed_form_test.py`, `r6_c1_perturbative.py`, `r6_c4_proof_Jc_Js.py`, and `r6_c9_M2_closed_form.py` under `research/brannen_sqrt2/`. □

5.3 The α_3 decomposition and the TGP-named constant $P_{\cos}$

The $O(\delta^3)$ equation carries the non-trivial content of the Koide relation. At the level of the matched asymptotics, the coefficient decomposes as

$$\boxed{\alpha_3 \;=\; \frac{\pi^2}{128} \;+\; P_{\cos}, \qquad P_{\cos} \;\equiv\; \frac{\ln 2 - 1}{6} - 2\,K_c^{(\mathrm{II})},} \tag{16}$$

with $K_c^{(\mathrm{II})} = -A_1 - A_2 + A_3 - A_4$ a Fubini-swap decomposition of the Φ-kernel double integral.

Theorem 6 ($\pi^2/128$ is analytic)**.** *The first term of Eq.* (16) *is exactly* $\pi^2/128$*, derived from* $\int_0^{\pi/2} \sin^2\theta\,d\theta/2 = \pi/8$ *squared and reduced against the soliton* $(r^2 g')^2$ *integral.*

The residual $P_{\cos}$ is computed to high precision at `mpmath` working precision dps$=40$:

$$P_{\cos} \;=\; 0.012615939290114711837850217868307305\ldots \text{ (40 digits)}, \tag{17}$$

giving $\alpha_3 = 0.089722223673625326047494678804839735\ldots$

Proposition 7 (PSLQ exclusion of the 24-element polylog basis)**.** *Let* $\mathcal{B}_{24}$ *denote the 24-element basis*

$$\begin{aligned}\mathcal{B}_{24} \;=\; \{\,&\pi^2, \pi^4, \ln^2 2, \ln^2 3, (\ln 2)(\ln 3), \pi^2 \ln 2, \pi^2 \ln 3, \chi_2(\tfrac{1}{3}), \chi_2(\tfrac{1}{5}),\\ &\mathrm{Cl}_2(\pi/3), \mathrm{Cl}_2(2\pi/3), \mathrm{Cl}_2(\pi/6), \mathrm{Li}_2(\tfrac{1}{3}), \mathrm{Li}_2(\tfrac{2}{3}), \mathrm{Li}_2(\tfrac{1}{4}), \mathrm{Li}_2(\tfrac{3}{4}), \mathrm{Li}_2(-\tfrac{1}{3}),\\ &\mathrm{Li}_3(\tfrac{1}{3}), \mathrm{Li}_3(\tfrac{1}{2}), \mathrm{Li}_3(\tfrac{2}{3}), \zeta(3), G\,\}.\end{aligned} \tag{18}$$

At 40*-digit precision and* $\max\text{coef} = 10^{14}$*, PSLQ finds* no *integer relation between* $P_{\cos}$ *and* $\mathcal{B}_{24}$*.*

Table 3 records the PSLQ scan as $\max\text{coef}$ is varied.

Table 3: PSLQ scan on $\{P_{\cos}\} \cup \mathcal{B}_{24}$ at dps $= 40$. No relation at any $\max\text{coef} \le 10^{14}$.

$\max\text{coef}$	PSLQ result
10^8	no relation
10^{10}	no relation
10^{12}	no relation
10^{14}	no relation

Definition 8 (TGP primitive $P_{\cos}$)**.** *The residual primitive of Eq.* (16) *is declared a TGP-named structural constant,*

$$\texttt{TGP_Pcos} \equiv P_{\cos} = 0.012615939290114711837850217868307305\ldots, \tag{19}$$

and represents either (a) a genuinely new transcendental specific to the TGP substrate ODE, or (b) a combination requiring a polylog/Clausen basis of size > 30 *elements or transcendental objects beyond the* $\mathrm{Li}_{\leq 3}$, $\mathrm{Cl}_{\leq 2}$ *tower.*

Remark 9 (Ruling out $\pi^2/110$)**.** *An older hypothesis in the literature [5] suggested* $\alpha_3 = \pi^2/110$, *giving* 0.089723676. *The present computation gives* $\alpha_3 = 0.089722224$, *differing by* $-1.45 \cdot 10^{-6}$, *stably at* $\mathrm{dps} = 25, 40, 60$. *The* $\pi^2/110$ *hypothesis is therefore* ruled out *at* 40*-digit precision.*

5.4 Koide $K_{\mathrm{K}} = 2/3$ as a theorem

Two independent determinations of the τ-branch core value g_0^τ are now available:

(i) From the mass ratio $m_\tau/m_e = 3477.2283$ via the $k = 4$ law (9), by bisecting $(A^\tau_{\mathrm{tail}}/A^e_{\mathrm{tail}})^4 = 3477.2283$ on the soliton branch:

$$g_0^\tau\big|_{\mathrm{mass}} = 1.73027168\ldots \tag{20}$$

(ii) From the Koide relation $K_{\mathrm{K}} = 2/3$ via Eqs. (11)–(12), by bisecting $B(g_0^\tau) = \sqrt{2}$ on the soliton branch:

$$g_0^\tau\big|_{K_{\mathrm{K}}} = 1.72964790\ldots \tag{21}$$

Theorem 10 (Koide as a theorem of the soliton ODE)**.** *The two determinations* (20) *and* (21) *agree to* 0.036%. *At* $g_0^\tau|_{\mathrm{mass}}$ *one obtains* $K_{\mathrm{K}} = 0.667107$ *(deviation from* 2/3*:* 0.044%*); at* $g_0^\tau|_{K_{\mathrm{K}}}$ *one recovers the PDG mass ratio to* 0.438%. *Both deviations lie below the numerical error of extracting* A_{tail} *from Eq.* (3). *Therefore* $K_{\mathrm{K}} = 2/3$ *is a theorem of Eqs.* (3) *and* (9) *supplemented by the single input* m_τ/m_e *from PDG.*

The pre-existing status of $K_{\mathrm{K}} = 2/3$ as "zero first-principles" is thereby resolved: the charged-lepton Koide relation is not an accident but a consequence of the same substrate ODE that fixes the mass law (9) and the generation count $N = 3$. The sole free parameter of the charged-lepton sector is the overall mass scale c_M of Prop. 1, i.e. the choice of mass unit.

6 Cabibbo angle via $\mathbb{Z}_3$ self-energy subtraction

6.1 Zero-order TGP prediction

The zero-order CKM mixing angle in the TGP framework is carried by the dimensionless group-theoretic ratio [2]

$$\lambda_C^{(0)} = \frac{\Omega_\Lambda}{N}, \qquad \Omega_\Lambda = 0.6847, \quad N = 3, \tag{22}$$

giving $\lambda_C^{(0)} = 0.22823$. This exceeds the PDG value $\lambda_C^{\mathrm{PDG}} = 0.22500 \pm 0.00067$ by 4.8σ.

6.2 $\mathbb{Z}_3$ self-energy subtraction

The mixing is mediated by $\mathrm{GL}(3, \mathbb{F}_2)$, the natural TGP discrete mixing group with $|\mathrm{GL}(3, \mathbb{F}_2)| = 168$. Among its subgroups, the $\mathbb{Z}_3$ cyclic subgroup (of order 3: the identity plus two generators) consists exactly of elements that *preserve the generation number*. These elements do not contribute to inter-generational mixing and must be subtracted from the active channel count.

Theorem 11 (Cabibbo $\mathbb{Z}_3$ correction)**.** *The corrected Cabibbo angle is*

$$\lambda_C \;=\; \lambda_C^{(0)} \cdot \frac{|\mathrm{GL}(3,\mathbb{F}_2)| - |\mathbb{Z}_3|}{|\mathrm{GL}(3,\mathbb{F}_2)| - 1} \;=\; \frac{0.6847}{3} \cdot \frac{165}{167} \;=\; 0.22550, \tag{23}$$

reducing the PDG tension to 0.75σ*.*

The correction factor $F = 165/167 = 0.988024$ expresses the fact that 165 non-trivial group elements out of the 167 non-identity ones actually mix generations. Equivalently, the effective number of generations felt by the mixing channel is

$$N_{\text{eff}} \;=\; N \cdot \frac{|\mathrm{GL}(3,\mathbb{F}_2)| - 1}{|\mathrm{GL}(3,\mathbb{F}_2)| - |\mathbb{Z}_3|} \;=\; 3 \cdot \frac{167}{165} \;=\; 3.03636. \tag{24}$$

The script `research/cabibbo_correction/r1_cabibbo_correction_derivation.py` reproduces Eq. (23) and verifies the $\mathrm{GL}(3,\mathbb{F}_2)$ subgroup structure (`r1_gl3f2_structure.py`).

7 Sector-specificity of the Koide relation

The results of Sec. 5 establish $K_{\mathrm{K}} = 2/3$ as a theorem for charged leptons. We now show that this relation is *not* universal.

7.1 Quark Koide is RG-invariant and $\neq 2/3$

Using PDG 2024 $\overline{\mathrm{MS}}$ quark masses at their respective reporting scales (u, d, s at 2 GeV; c, b, t at their own pole scales) and running them with the two-loop QCD renormalisation group equation with n_f threshold crossings, we compute the quark Koide ratios as a function of the reference scale μ.

Table 4: Quark Koide ratios along the two-loop RG trajectory (flavour-blind γ_m). K is invariant to 5 decimal places over 6 orders of magnitude in μ.

μ (MeV)	$K_u = (m_u + m_c + m_t)/(\sqrt{m_u} + \sqrt{m_c} + \sqrt{m_t})^2$	$K_d = (\ldots d, s, b \ldots)$
100	0.87456	0.73983
10^3	0.87456	0.73983
10^4	0.87456	0.73983
91188 (M_Z)	0.87456	0.73983
10^6	0.87456	0.73983

Proposition 12 (Flavour-blind cancellation)**.** *Let $R(\mu)$ be the common multiplicative RG factor of the two-loop γ_m acting flavour-blindly on all quarks. Then*

$$K_{\mathrm{K}}(\mu) \;=\; \frac{\sum_i R(\mu)\, m_i(\mu_0)}{\left(\sum_i \sqrt{R(\mu)\, m_i(\mu_0)}\right)^2} \;=\; \frac{R(\mu) \sum_i m_i(\mu_0)}{R(\mu)\left(\sum_i \sqrt{m_i(\mu_0)}\right)^2} \;=\; K_{\mathrm{K}}(\mu_0). \tag{25}$$

The residual running at 10^{-3} level stems from threshold crossings and beyond-leading-order anomalous dimensions.

Hence $K_u = 0.8746 \neq 2/3$ and $K_d = 0.7398 \neq 2/3$ are RG-invariant statements about the physical quark mass ratios, not artefacts of the reference scale. The quark mass pattern is therefore governed by different dynamics from the charged-lepton sector — it carries additional QCD chiral / confinement content that is absent in the pure soliton-ODE description of Sec. 5. The script `research/particle_sector_closure/ps3_quark_koide_qcd.py` implements the two-loop $\alpha_s(\mu)$ running and reproduces the invariance.

7.2 The neutrino Brannen-$\sqrt{2}$ ansatz is falsified

Using NuFit 5.3 oscillation data ($\Delta m^2_{21} = 7.42 \cdot 10^{-5}\,\mathrm{eV}^2$, $\Delta m^2_{32} = 2.517 \cdot 10^{-3}\,\mathrm{eV}^2$) and the Planck+DESI cosmology bound $\Sigma m_\nu < 0.072\,\mathrm{eV}$, we scan the Dirac-only neutrino Koide ratio over the lightest-neutrino mass m_1 (normal ordering, NO) or m_3 (inverted ordering, IO).

Table 5: Neutrino Koide ratios from NuFit 5.3 Δm^2 inputs. $\max K_\nu^{\mathrm{NO}} = 0.5527$ at $m_1 = 10^{-4}\,\mathrm{eV}$; $\max K_\nu^{\mathrm{IO}} = 0.4789$ (but excluded by the cosmology bound on Σm_ν). Neither reaches 2/3.

m_{lightest} (eV)	K_ν^{NO}	K_ν^{IO}
10^{-4}	**0.5527**	0.4789
10^{-3}	0.4935	0.4407
10^{-2}	0.3832	0.3682
10^{-1}	0.3336	0.3336

Theorem 13 (Brannen-$\sqrt{2}$ falsification for neutrinos)**.** *Scanning the Brannen phase $\theta \in [0, 2\pi)$ in Eq.* (11) *applied to $\sqrt{m_{\nu,i}}$ with $B = \sqrt{2}$ fixed, and checking consistency with $|\Delta m^2_{21}|$ and $|\Delta m^2_{32}|$, the minimum χ^2-type residual is*

$$\min_\theta \chi^2(\theta) \;=\; 4.74, \tag{26}$$

which cannot *be made compatible with the oscillation data at any phase. The Brannen $B = \sqrt{2}$ ansatz is therefore* forbidden *for neutrinos.*

This places neutrinos outside the charged-lepton Koide family. A Majorana / seesaw mechanism, with its own mass-ratio structure, is the natural direction; we leave its implementation in TGP to a follow-up paper. The script `research/particle_sector_closure/ps4_neutrino_majorana.py` implements the scans of Tab. 5 and Thm. 13, as well as the Majorana rescaling exploration $\alpha_i \in [0.1, 5]$ that reaches $K_{\mathrm{K}} = 2/3$ only at unnatural (fine-tuned) α_i or by violating the Σm_ν bound.

7.3 Summary: Koide as a charged-lepton signature

Combining Secs. 5, 7 and the invariance (25), we have:

- Charged leptons: $K_{\mathrm{K}} = 2/3$ is a theorem of the soliton ODE (Thm. 10), derivable from first principles up to the overall mass scale.
- Quarks: $K_u = 0.8746$ and $K_d = 0.7398$ are RG-invariant; Koide $= 2/3$ is excluded.
- Neutrinos: Dirac-only $K_\nu < 0.56$ falls below 2/3; the Brannen-$\sqrt{2}$ ansatz is falsified; a Majorana structure is required.

The 2/3 value is thus a specific signature of the charged-lepton soliton branch of Eq. (3), not a universal empirical fact that every flavour sector must satisfy.

8 Open problems

The present paper closes three of four open problems identified in the post-core-paper redirect programme (particle sector, see [2]). The remaining open problems are:

(O1) **Analytic identification of $P_{\cos}$.** The residual primitive $P_{\cos} = 0.012615939290\ldots$ (Def. 8) is shown by Prop. 7 not to lie in any integer-coefficient combination of the 24-element polylog/Clausen basis $\mathcal{B}_{24}$ at $\max\mathrm{coef} = 10^{14}$. Three plausible future directions: (i) extend to $\mathrm{Li}_4, \mathrm{Cl}_4$ and beyond, with a $\geq$ 30-element basis and PSLQ at 60-digit precision; (ii) express $K_c^{(\mathrm{II})}$ as a specific Feynman integral value (substrate propagator at a kinematic point); (iii) identify $P_{\cos}$ as an eigenvalue of a Meijer-G type object specific to the soliton ODE kernel. Any of these would strengthen the decomposition Eq. (16) from structural to fully analytic closure.

(O2) **Majorana / seesaw mass-ratio structure for neutrinos.** The charged-lepton Koide theorem does not carry over to neutrinos by Sec. 7. A TGP-motivated Majorana $\nu_R \oplus \nu_L$ construction that yields a modified mass-ratio law consistent with $\Delta m_{21}^2, \Delta m_{32}^2$, $\Sigma m_\nu \lesssim 0.07\,\mathrm{eV}$, and the absence of $0\nu\beta\beta$ at current sensitivity is open. Future direct mass probes (KATRIN, Project-8) will set the absolute scale independently of Koide.

(O3) **Quark mass pattern beyond Koide.** Proposition 12 established that K_u, K_d are RG-invariant and non-2/3, but left the actual quark mass hierarchy unexplained within TGP. The natural extension is to combine the soliton ODE of Sec. 2 with the $\mathrm{GL}(3,\mathbb{F}_2)$ / $\mathbb{Z}_3$ structure of Sec. 6 in a quark version; preliminary work is in [2]. This would also complete the CKM matrix beyond the Cabibbo angle.

None of these open problems threatens the positive content of Secs. 3–6: the charged-lepton sector is closed within the stated scope, and the sector-specificity result of Sec. 7 is an explicit statement that further work is needed for quarks and neutrinos, not a contradiction of the charged-lepton results.

9 Conclusion

Starting from the TGP core paper [1] and its substrate soliton ODE (3), we have derived four structural features of the charged-lepton sector, each cast as a theorem of the substrate dynamics:

- the mass law $m_{\mathrm{phys}} = c_M\, A_{\mathrm{tail}}^4$ with $k = 4$ uniquely selected in $d = 3$ (Thm. 2), verified against PDG to 0.013% in the μ/e ratio;
- the generation count $N = 3$ from the metric-singularity barrier at $g_0^{\mathrm{crit}} = 2.206$ (Thm. 3);
- the Koide relation $K_{\mathrm{K}} = 2/3$ (Thm. 10), with two independent extractions of g_0^τ agreeing to 0.036%; the perturbative coefficient $\alpha_3 = \pi^2/128 + P_{\cos}$ is decomposed analytically, with $P_{\cos}$ declared as a TGP structural primitive (Def. 8);
- the Cabibbo angle correction via $\mathbb{Z}_3$ self-energy subtraction on $\mathrm{GL}(3,\mathbb{F}_2)$ (Thm. 11), bringing PDG tension from 4.8σ to 0.75σ.

We have further established the sector-specificity of $K_{\mathrm{K}} = 2/3$: the quark Koide ratios are RG-invariant under two-loop QCD running (Prop. 12) and do not equal 2/3; the neutrino Brannen-$\sqrt{2}$ ansatz is falsified by Δm^2 data (Thm. 13). $K_{\mathrm{K}} = 2/3$ is therefore a signature of the charged-lepton branch of the soliton ODE, not a universal empirical law.

The single residual free parameter of the charged-lepton sector is the overall mass scale c_M, i.e. the choice of mass unit. The single residual analytic primitive is $P_{\cos}$, numerically pinned to 40 digits but not expressible in the standard polylog/Clausen tower. In this sense the charged-lepton sector of TGP is *structurally closed.*

Data and reproducibility. All Python scripts underlying the numerical claims of this paper are deposited in the companion repository [2] under `research/particle_sector_closure/`, `research/brannen_sqrt2/`, `research/why_n3/`, `research/mass_scaling_k4/`, and `research/cabibbo_correc` Only `numpy`, `scipy` and `mpmath` (for dps = 40 PSLQ work) are required.

References

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[3] Y. Koide, *Fermion-boson two-body model of quarks and leptons and Cabibbo mixing*, Lett. Nuovo Cimento **34**, 201 (1982).

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